
PDS background activity and dead time

Current estimate, simulation chain, and remaining uncertainty

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Candidate activity is not board acceptance

NOMINAL INTRINSIC-LAr RESULT

A VD cathode channel
sees 3.59 times the HD
candidate activity.

The mixture is dominated by atmospheric
 ^{39}Ar ; at the same 0.7-PE threshold:
87.47 kHz/ch for VD cathode and 24.34 kHz/ch
for HD.

01 LArSoft output

Candidate times and rates.

02 Firmware estimate

Channel-local busy gate.

03 Not yet modeled

Shared board transport.

Assumptions: source, threshold, and pulse shape

dunesw v10_22_00d01 | LArSoft v10_22_00 | art 3.14.04

Item	Condition used for the headline result
Source	Uniform intrinsic-LAr mixture: atmospheric ^{39}Ar at 0.964 Bq/kg, plus released ^{85}Kr , ^{42}Ar , and ^{42}K ; eight independent shards per detector.
Optical threshold	Every local candidate above 0.7 PE is counted. There is no 10-MeV trigger cut. Energy is retained only for classification and auditing.
Pulse model	The 16 Waffles NP02 cathode C-channel large_pulses_zero_subtract waveforms form one common SPE-shape set for HD and VD.

MODEL ASSUMPTION

The common C-channel pulse set isolates geometry and activity

It is not evidence that HD and VD have identical gain, noise, PDE, or technology-specific SPE shapes.

What we simulated—in chronological order

Step	Operation	Meaning
1	Physical source	LArSoft draws radioactive decays during a known time interval, or one gamma from a controlled energy/face cell.
2	Full geometry	Geant4 transports particles through the complete HD-v6 or VD-v7 detector geometry.
3	Optical response	Energy deposited in liquid argon produces light and the detector-specific photon response.
4	Common pulse model	Detected light is shaped with the same 16 Waffles NP02 cathode-C templates in both technologies.
5	Candidate + firmware	A strict > 0.7 -PE crossing forms a candidate; XC fixed-record busy time is applied afterward, channel by channel.

Physical windows give Hz; stratified particles give response.

A response matrix becomes a rate only after applying the physical source weights.

Full geometry and channel denominator

Item	Condition used for the headline result
FD-HD full detector	LArSoft dune10kt_v6_full: 6,000 APA-integrated optical hardware channels; HD is the reference.
FD-VD full detector	LArSoft dunevd10kt_3view_30deg_v7_full: 1,344 channels—640 cathode, 640 long-wall, and 64 short-wall.
Optical response	Geometry and photon transport remain technology-specific. Only the Waffles C-channel SPE pulse-shape set is common.
Denominator	All hardware channels in the relevant population, including quiet channels; units are candidates per channel per second.

DEFINITION

Candidate $\neq$ **accepted frame**

A candidate is a threshold-crossing opportunity on one channel. Firmware acceptance is evaluated afterward.

What drives the nominal intrinsic-LAr rate

Component	Generator content	Share of generated primaries
^{39}Ar	Atmospheric argon at 0.964 Bq/kg	89.4%
^{85}Kr	Released intrinsic-LAr normalization	10.6%
^{42}Ar , ^{42}K	Released intrinsic-LAr normalizations	< 0.02% combined
Other families	Detector materials, radon chains, and external sources	Not in this baseline

CORRECTION

The 24.34/87.47-kHz numbers are not Ar-39-only

The source labels are merged before optical simulation. We can state the generated-primary mix, but cannot assign the final candidates to one isotope from this sample.

Background scope: what is not closed

Family	Current content	Status
External γ	Cavern-wall and foam; cryostat- and cavern-neutron-induced γ generators	One physical window per family; pilot only
Neutrons / captures	Actual cavern neutrons, capture timing and secondary γ , alpha-capture γ	Not closed; excluded
Remaining radon	^{214}Bi and the missing $^{219}\text{Rn}/^{220}\text{Rn}$ chain rows	Incomplete; excluded
Electronics	Dark counts, line/electronics noise, afterpulsing, and cross-talk	No validated physical-rate component

NO TOTAL-BACKGROUND CLAIM

The gamma extension is a first physical-window estimate

Its candidate ticks are overlaid with the nominal stream before XC busy time. More gamma windows and the missing families are still required.

The matched gamma-response pilot is complete

Conditional detector response—physical gamma weights are not yet applied

Detector	Incident γ	Active-LAr deposit	> 0.7-PE response	Probability
FD-HD	10,000	2,458	2,358	23.58%
FD-VD	10,000	1,868	1,782	17.82%

EVENT RESPONSE

$$\text{VD} / \text{HD} = 0.756$$

This compares the same artificial proposal: ten energy bins, five entry faces, and 200 independent gammas per cell.

CHANNEL-NORMALIZED YIELD

$$\text{VD} / \text{HD} = 0.912$$

Candidate counts divided by 1,344 VD or 6,000 HD channels and by the same number of injected gammas.

HD low-z gave zero active-LAr deposits despite verified inward directions; retain as a pilot observation and validate more densely.

What is—and is not—a background-rate result

Quantity	Status	Correct interpretation
Nominal intrinsic LAr	Rate-ready	^{39}Ar at 0.964 Bq/kg plus ^{85}Kr , ^{42}Ar , and ^{42}K ; eight windows per detector.
External- γ response matrix	Complete	Conditional efficiency versus energy and face; not Hz/channel.
Physical external- γ rate	Pilot complete	Released physical generators in full geometry; one window per source family.
Total background stream	Not closed	Requires all source families overlaid in time before XC and shared-board back-pressure.

Two labels are kept separate: nominal intrinsic LAr and intrinsic LAr + gamma pilot.
Neither is yet a complete board-level background model.

Physical gamma pilot: first absolute rates

One full-geometry physical window per source family; strict > 0.7 PE

Population	Cavern wall kHz per channel	Foam	Total gamma
HD APA-integrated	18.805	0.719	19.523
VD cathode	4.731	0.184	4.915
VD long wall	21.805	0.943	22.749
VD short wall	15.461	0.675	16.136

PILOT LIMIT

Zero observed is not zero expected

The cryostat-neutron- and cavern-neutron-gamma generators produced one primary each and no candidates in these windows. The deposited-energy audit also timed out, so the energy population is not yet validated.

Nominal intrinsic-LAr activity depends on position

Population	Mean kHz per channel	Min-max	Mean / HD
HD APA-integrated	24.34	6.68-44.77	1.00
VD cathode	87.47	58.99-107.03	3.59
VD long wall	30.49	20.03-40.16	1.25
VD short wall	22.05	16.21-27.69	0.91

READ THE RANGE

Mean $\neq$ maximum

Hottest VD cathode
channel: **107.03** kHz/ch.

NOMINAL RATIO

Cathode / HD = 3.59

87.47/24.34, before
firmware loss.

Candidate-level gamma overlay: provisional load

shards

Population	Intrinsic LAr candidate	Gamma kHz/ch	Combined	Accepted [kHz/ch]	
				1024	512
HD APA-integrated	24.338	19.523	43.855	21.030	26.152
VD cathode	87.467	4.924	92.383	34.534	48.297
VD long wall	30.490	22.794	53.273	25.020	32.167
VD short wall	22.050	16.136	38.181	21.264	26.238

WHAT THE OVERLAY ADDS

Cross-source candidate-level pileup

Busy rejection is computed after merging timestamps: HD 52.05% → 40.37%, and VD cathode 62.62% → 47.72%, for 1024 → 512 samples.

VD gamma candidates are clipped from the native 8.57-ms window to the nominal 8.50-ms window before overlay. Analog pileup and shared board back-pressure are not modeled.

The conversion to offered board traffic is explicit

$$W_{\text{rec}} = 8 + 7 \left\lceil \frac{N_s}{32} \right\rceil, \quad B_{\text{rec}} = 64W_{\text{rec}}, \quad T_{\text{board}} = N_{\text{ch}} R_{\text{cand/ch}} B_{\text{rec}}$$

Case	Channels	Rate/ch	Bits/record	Offered traffic
VD cathode, 512	32	87.47 kHz	7,680	21.5 Gbit/s
HD, 512	40	24.34 kHz	7,680	7.48 Gbit/s

512 $\rightarrow$ 120 64-bit words; 1024 $\rightarrow$ 232 words = 14,848 bits.

CHANNEL COUNT MATTERS

Homogeneous VD-cathode / HD board load = 2.87

This is an offered-load illustration, not an accepted-output prediction or a detector wiring claim.

Why $14 \times 512 \times 32$ is not yet throughput

BITS IN ONE BOARD-WIDE SET

A size, not a rate

$14 \text{ bit/sample} \times 512 \text{ sample/frame} \times$
 $32 \text{ ch} = 229,376 \text{ bit}.$

BITS PER SECOND

Multiply by activity

With the modeled record overhead:
 $32 \text{ ch} \times R_{\text{ch}} \times 7,680 \text{ bit/frame}.$

512 + 50% OVERLAP

Overlap changes acceptance opportunities, not record size

Each fragment is still 7,680 bits. The 256 overlapping samples are retransmitted unless firmware coalesces them: $T = R_{\text{accepted,overlap}} \times 7,680.$

What the present firmware replay actually does

Stage	Included here	Still separate
Candidate formation	LArSoft timestamp for each strict > 0.7 -PE episode	Continuous ADC overlay across source families
XC channel logic	Fixed busy gate: 1,037 cycles for 1024; 525 for 512	ADC-level XC FIR/CFD decision and efficiency
Window overlap	Record size is unchanged	512 + overlap acceptance/coalescing policy
Shared transport	None	Board map, FIFO, lanes, serializer, Hermes ready, and 10-Gbit/s back-pressure

Keep the evidence separate: the $> 89\%$ result is from a different XC injected-signal test at 1 PE and above; it is not an efficiency measured by this replay.

Hermes back-pressure changes acceptance

Intrinsic-LAr + one-window gamma pilot; homogeneous-board illustration

Sender input: $T_{\text{in}} = R_{\text{accepted, local}} B_{\text{record}}$; queue growth: $dQ/dt = \max(0, T_{\text{in}} - C_{\text{payload}})$, where $C_{\text{payload}} < 10 \text{ Gbit/s}$.

Population	512 demand $\rightarrow T_{\text{in}}$	1024 demand $\rightarrow T_{\text{in}}$	Ideal-link test
HD	13.47 $\rightarrow$ 8.03	26.05 $\rightarrow$ 12.49	1024 saturates
VD cathode	22.70 $\rightarrow$ 11.87	43.90 $\rightarrow$ 16.41	both saturate
VD long wall	13.09 $\rightarrow$ 7.91	25.31 $\rightarrow$ 11.89	1024 saturates
VD short wall	9.38 $\rightarrow$ 6.45	18.14 $\rightarrow$ 10.10	1024 at limit

All traffic values are Gbit/s for one homogeneous board and one 10-Gbit/s sender. “Demand” is before the local gate; T_{in} is after it.

CONSEQUENCE

The ideal 10-Gbit/s comparison is already unfavorable

All 1024-sample populations meet or exceed line rate; VD cathode also exceeds it at 512.

Usable payload is below 10 Gbit/s, so final loss requires Hermes ready, FIFO depth, and the real channel-to-sender map.

512 samples improves the channel-local estimate

Nominal intrinsic-LAr baseline; fixed-record busy gate applied independently per channel

Population	Accepted rate [kHz/ch]		Rejected while busy	
	1024	512	1024	512
HD APA-integrated	15.45	17.91	36.5%	26.4%
VD cathode	33.62	46.53	61.6%	46.8%
VD long wall	18.78	22.27	38.4%	27.0%
VD short wall	15.47	17.76	29.9%	19.5%

SUPPORTED

The direction of the effect

Shortening 1024 to 512 samples increases accepted candidate rate in every population tested.

NOT SUPPORTED YET

A board-level dead-time number

This replay has no shared FIFO, serializer, lane assignment, or Hermes ready/back-pressure.

The remaining uncertainty is architectural

WHAT WE KNOW

Detector-side activity

Nominal intrinsic-LAr rates and ranges, a one-window physical gamma pilot, VD populations, and the direction of the 512-sample effect.

WHAT IS REQUIRED

A board-aware replay

Map channel → board → sender; replay correlated times through FIFO, serializer, Hermes ready, and the 10-Gbit/s link.

REPORTING RULE

Keep the labels separate

Candidate activity, channel-local acceptance, and offered sender traffic are different quantities. Reserve *board throughput/dead time* for the mapped model.

What we can say today

DETECTOR ACTIVITY

The VD cathode is demanding

Intrinsic LAr: 87.47 versus HD 24.34 kHz/ch (3.59 $\times$). Gamma pilot: 92.38 versus 43.85 (2.11 $\times$).

RECORD LENGTH

512 samples helps

VD-cathode rejection: 61.6% $\rightarrow$ 46.8% in the baseline; 62.6% $\rightarrow$ 47.7% with the gamma pilot.

TRANSPORT LIMIT

One 10-Gbit/s link is tight

Gamma pilot after the local gate: VD cathode 11.87 Gbit/s at 512; 16.41 Gbit/s at 1024.

NEXT CLOSURE

The estimate remains provisional

Repeat gamma windows; add missing backgrounds; validate energy; then map boards, FIFO, lanes, serializer, and Hermes ready.